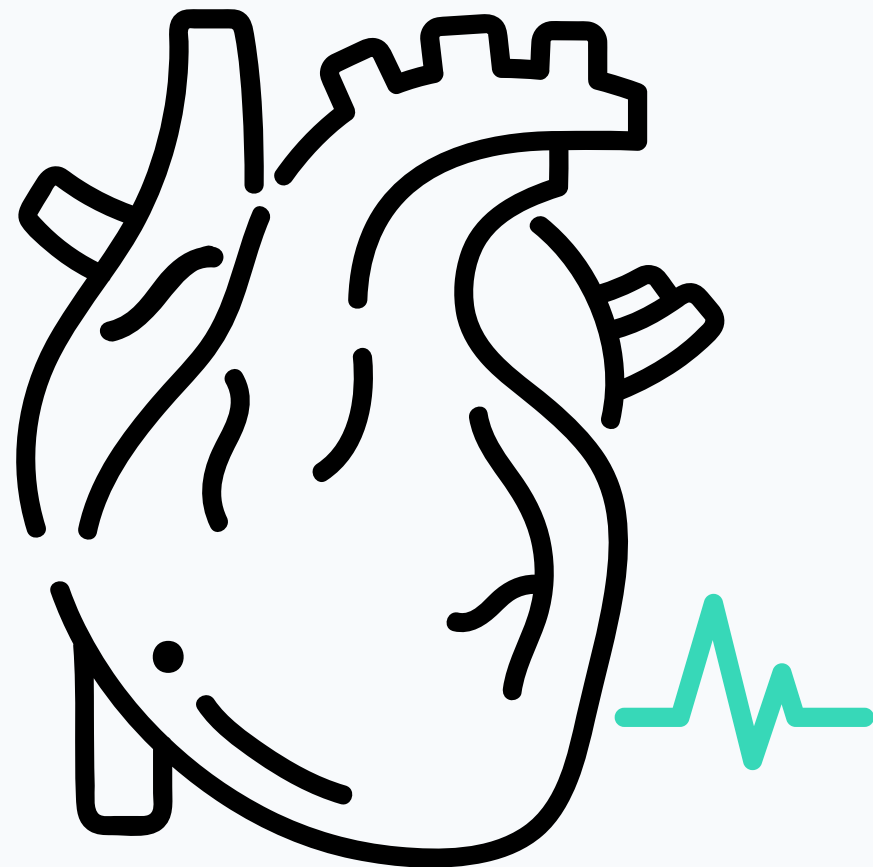


Heart Disease Risk Prediction using Machine Learning

Healthcare Analytics Project
by a Biotechnology Engineer

Python • Scikit-Learn • XGBoost • Healthcare Analytics

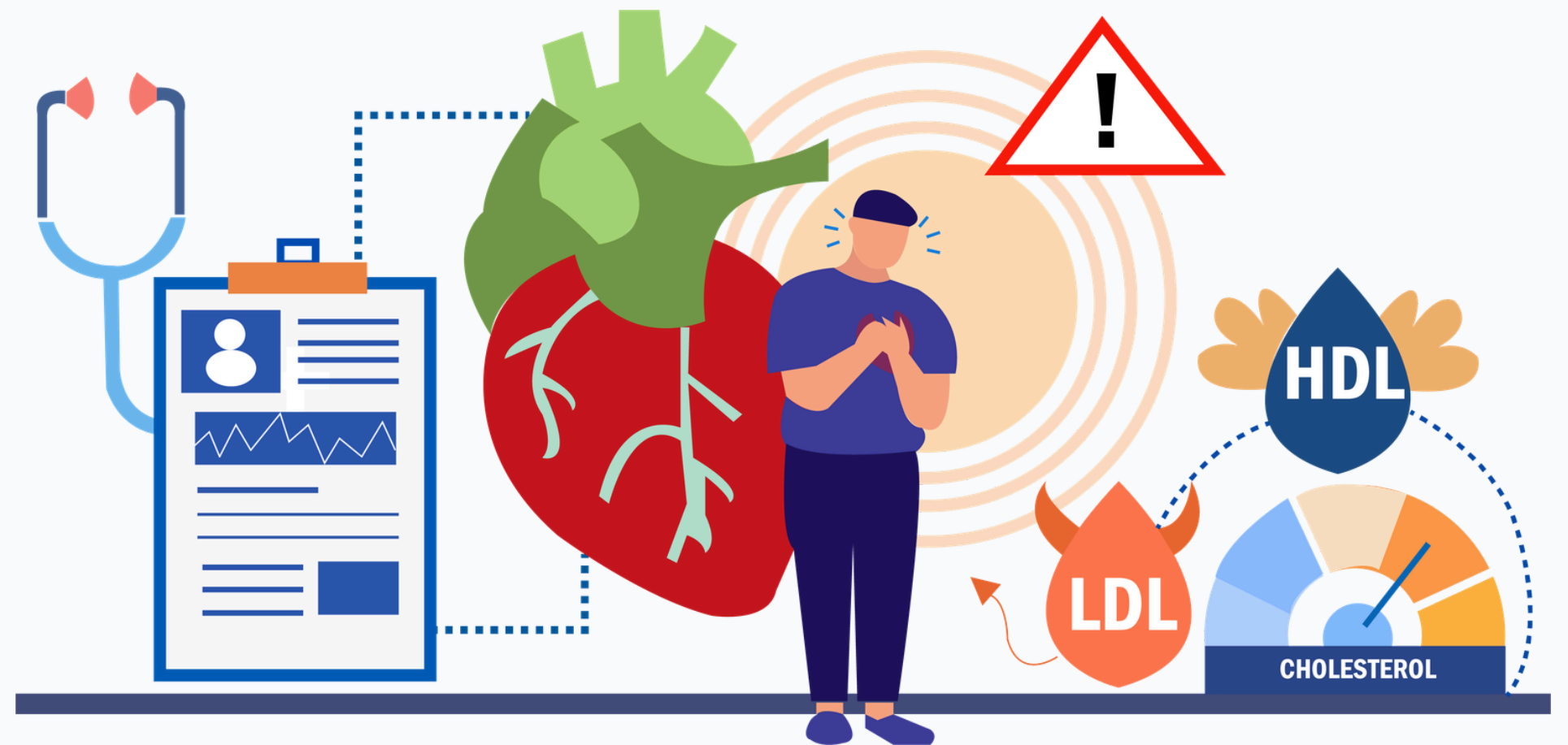


Can Machine Learning Help Detect Heart Disease?

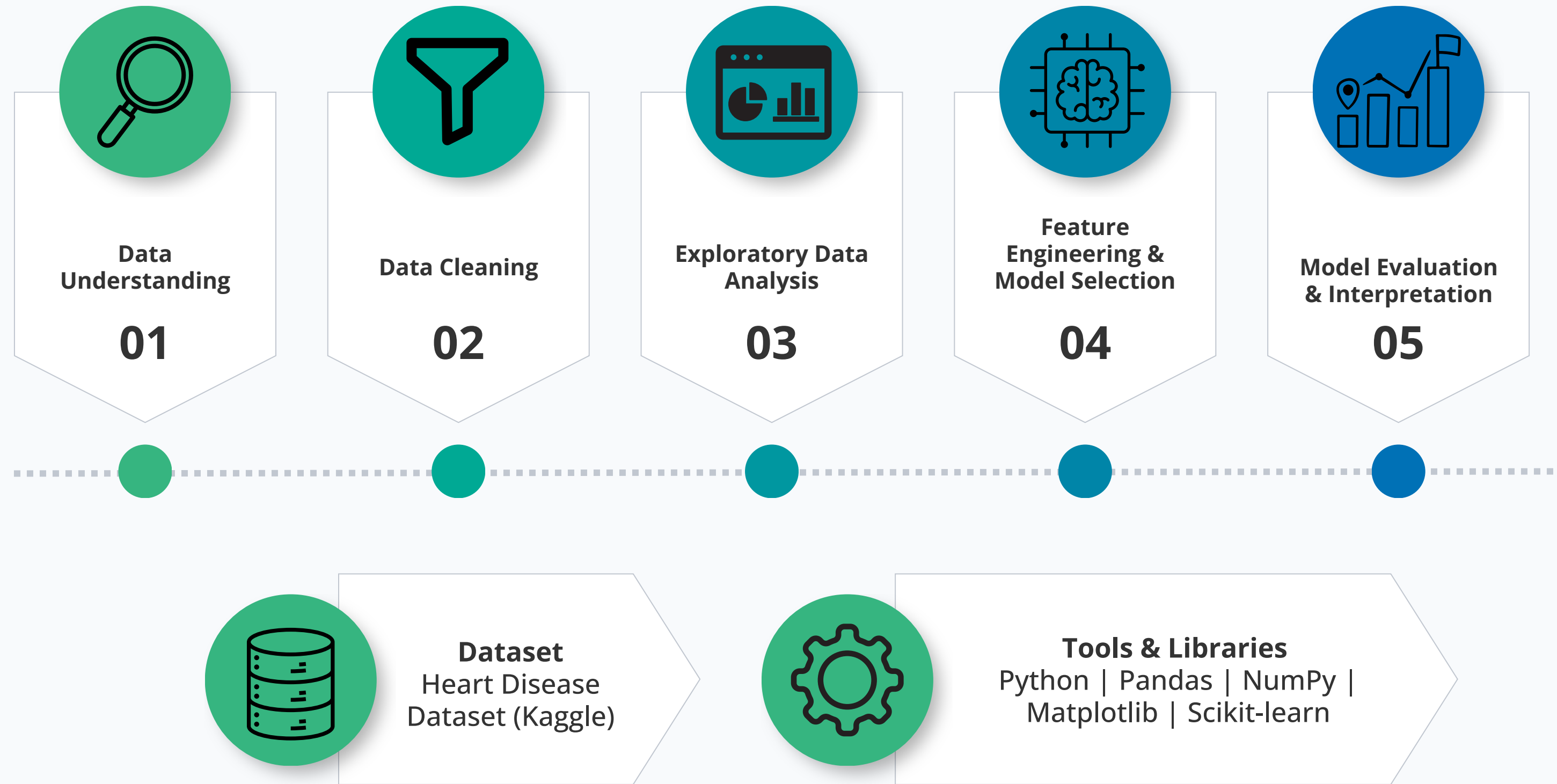
- Cardiovascular diseases remain one of the leading causes of death worldwide.
- Early detection of high-risk patients can improve clinical decision-making and reduce severe outcomes.
- This project explores how machine learning models can support heart disease prediction using clinical patient data while balancing predictive performance and interpretability.

Main Objective

Reduce false negatives while maintaining strong classification performance.



Project Workflow

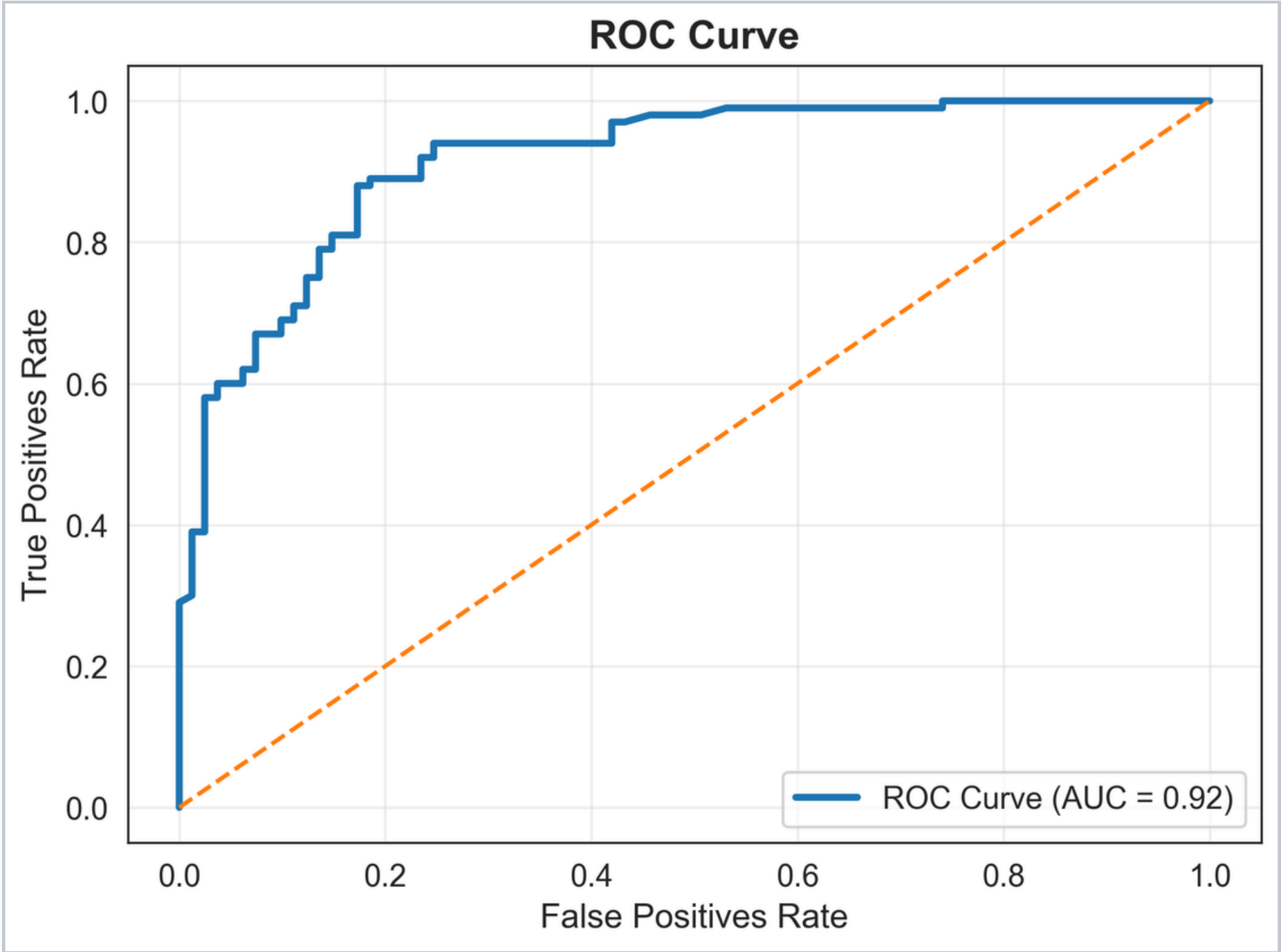


Model Performance

Model	ROC-AUC	Recall	F1-Score	Precision
Logistic Regression	0.92	0.88	0.87	0.86
Random Forest	0.89	0.85	0.83	0.81
XGBoost	0.82	0.89	0.78	0.69
Decision Tree	0.76	0.75	0.76	0.78

Logistic Regression achieved the best balance between predictive performance and interpretability.

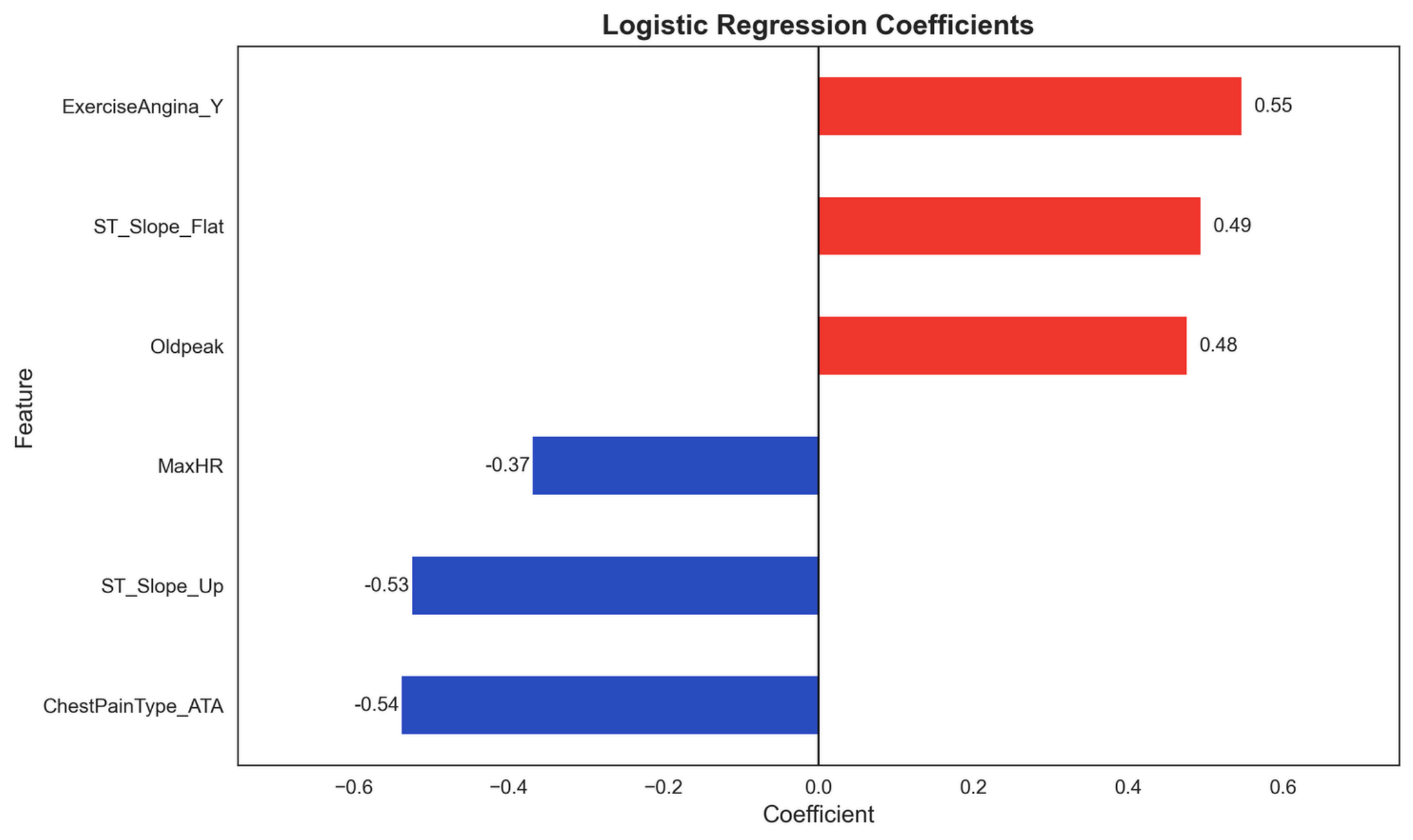
Balancing Sensitivity and False Negatives



Threshold	False Negative	Recall
0.5	12	0.88
0.3	7	0.93

- Reducing false negatives is critical in healthcare applications.
- Threshold optimization improved sensitivity, while the default threshold preserved a better balance between recall and precision.

Key Predictors of Heart Disease



- Exercise-induced angina showed the strongest positive association with heart disease risk.
- ST depression (Oldpeak) and flat ST slope were also linked to higher cardiovascular risk.
- Higher MaxHR and upsloping ST segments showed protective associations.

Key Insights

- **Logistic Regression achieved strong predictive performance while maintaining clinical interpretability.**
- **Exercise-induced angina and ST-segment abnormalities were strongly associated with cardiovascular risk.**
- **Threshold optimization demonstrated the importance of balancing sensitivity and precision in healthcare applications.**
- **The project combined biotechnology and data science to explore clinically meaningful predictive modeling.**



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 ..		
 data	TomasBarria	2 days ago
 images	Update	4 hours ago
 notebook	Update notebook	15 minutes ago
 .gitignore	Update	12 minutes ago
 README.md	Update README.md	4 hours ago
 requirements.txt	TomasBarria	2 days ago

README.md

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Python

3.13

Pandas

Data Analysis

Scikit-Learn

Machine Learning

XGBoost

Boosting

Domain

Healthcare

- ✓ End-to-end ML workflow
- ✓ Healthcare analytics
- ✓ Model evaluation & interpretation
- ✓ Logistic Regression + XGBoost

Combining Biotechnology and Data Science to explore healthcare insights

Full Project available on GitHub

Open to Data Analytics & Healthcare Analytics opportunities

